

MACD Divergence

Indicator & Strategy Guide

A comprehensive guide to detecting and trading
MACD divergences across multiple platforms

Platforms: MetaTrader 5 | TradingView | Python

Author: Community

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1. Introduction to MACD Divergence

MACD (Moving Average Convergence Divergence) is one of the most popular technical indicators used by traders worldwide. It measures the relationship between two exponential moving averages (EMAs) of a security's price.

A divergence occurs when the price of an asset moves in the opposite direction of a technical indicator. In the context of MACD, a divergence happens when:

- Price makes a new high but MACD does not (bearish divergence)
- Price makes a new low but MACD does not (bullish divergence)

Divergences are powerful signals because they often precede trend reversals. They indicate that the momentum behind a price move is weakening, even if the price itself is still trending.

Why MACD Divergences Matter

MACD divergences are particularly valuable because:

- They provide early warning of potential trend reversals
- They can confirm or contradict other technical signals
- They work across multiple timeframes (intraday, daily, weekly)
- They are relatively easy to identify visually

2. Divergence Types

This indicator detects four types of divergences, classified by their direction and strength:

2.1 Classical Bullish Divergence

Signal: Potential bottom / reversal to the upside

- Price makes a LOWER LOW
- MACD makes a HIGHER LOW
- Interpretation: Selling pressure is weakening

2.2 Reverse Bullish Divergence

Signal: Trend continuation to the upside

- Price makes a HIGHER LOW
- MACD makes a LOWER LOW
- Interpretation: Temporary pullback in uptrend

2.3 Classical Bearish Divergence

Signal: Potential top / reversal to the downside

- Price makes a HIGHER HIGH
- MACD makes a LOWER HIGH
- Interpretation: Buying pressure is weakening

2.4 Reverse Bearish Divergence

Signal: Trend continuation to the downside

- Price makes a LOWER HIGH
- MACD makes a HIGHER HIGH
- Interpretation: Temporary bounce in downtrend

3. Detection Algorithm

The detection algorithm uses a multi-step process to identify reliable divergences:

Step 1: MACD Calculation

```
EMA_fast = EMA(close, 12)
EMA_slow = EMA(close, 26)
MACD = EMA_fast - EMA_slow
Signal = EMA(MACD, 9)
Histogram = MACD - Signal
```

Step 2: Extrema Detection (3-Condition Check)

A bar is identified as a local trough or peak on the MACD line when:

- Current value $\leq$ previous value (non-strict)
- Current value $<$ two-bars-ago value (strict)
- Current value $<$ next-bar value (strict)

Step 3: Historical Search (4-Condition Check)

To find the last significant trough/peak, the algorithm searches backwards:

- First checks the Signal line with 4 conditions ($\leq$ and $<$ comparisons)
- Then verifies the MACD line has a corresponding trough/peak
- Search range: up to 5 bars back from current detection

Step 4: Divergence Comparison

Compare current and last extrema:

- Bullish: Compare lows of price and MACD
- Bearish: Compare highs of price and MACD
- Classical: Price and MACD move in opposite directions
- Reverse: Price and MACD move in same direction

4. Indicator Parameters

MACD Settings

- Fast Length: 12 (default) - Period for fast EMA
- Slow Length: 26 (default) - Period for slow EMA
- Signal Smoothing: 9 (default) - Period for signal line

Display Settings

- Draw Indicator Trend Lines: Show/hide trendlines on MACD panel
- Draw Price Trend Lines: Show/hide trendlines on price chart
- Display Alert: Enable/disable alerts on divergence detection

5. Trading Strategy

Entry Rules

LONG Entry:

- Detect Classical Bullish Divergence
- Enter at open of next bar after signal
- Optional: Close existing short position

SHORT Entry:

- Detect Classical Bearish Divergence
- Enter at open of next bar after signal
- Optional: Close existing long position

Exit Rules

- Stop Loss: Below recent swing low (long) or above swing high (short)
- Take Profit: Based on risk:reward ratio (default 2:1)
- Opposite Signal: Close position on opposite divergence

6. Risk Management

Position Sizing

The backtester uses risk-based position sizing:

$\text{Risk Amount} = \text{Capital} \times \text{Risk Per Trade}$

$\text{Risk \%} = |\text{Entry} - \text{SL}| / \text{Entry}$

$\text{Position Size} = \text{Risk Amount} / \text{Risk \%}$

Stop Loss Methods

- Swing: Use recent swing low/high (default)
- ATR: Based on Average True Range
- Fixed: Fixed percentage from entry

Take Profit Methods

- Risk:Reward: Multiplier of risk (default 2x)
- ATR: Based on Average True Range
- Opposite: Exit on opposite signal

7. Platform Implementations

MetaTrader 5 (MQL5)

- macd_divergence.mq5: Visual indicator with trendlines
- macd_divergence_ea.mq5: Automated trading EA
- Original implementation by Alain Verleyen

TradingView (Pine Script v5)

- macd_divergence.pine: Port of MQL5 logic
- Includes labels and trendlines
- Compatible with all TradingView assets

Python Backtester

- macd_divergence_backtest.py
- Uses yfinance for data
- Generates detailed performance reports
- Outputs equity curve and trade CSV

8. Backtesting Results

The backtester provides comprehensive metrics:

- Total Trades: Number of trades executed
- Win Rate: Percentage of profitable trades
- Profit Factor: Gross profit / gross loss
- Total Return: Overall account growth
- Max Drawdown: Largest peak-to-trough decline
- Sharpe Ratio: Risk-adjusted returns

Running a Backtest

```
python macd_divergence_backtest.py
```

```
# Or customize parameters:
bt = MACDDivergenceBacktest(
    symbol="EURUSD=X",
    start="2020-01-01",
    end="2025-01-01",
    interval="1d"
)
df, trades = bt.run_backtest()
metrics = bt.report(trades, df)
```

9. Best Practices

When to Use Divergences

- In trending markets (not ranging)
- On higher timeframes (4H, Daily, Weekly)
- Combined with other technical analysis
- With proper risk management

Common Mistakes to Avoid

- Trading divergences in isolation
- Ignoring the overall trend
- Using too tight stop losses
- Overtrading on every divergence signal

Optimization Tips

- Test different MACD parameters for your asset
- Adjust lookback periods for different timeframes
- Combine with volume analysis
- Use multiple timeframe analysis

10. References

Technical Analysis Resources:

- MACD - Investopedia
- Divergence Trading - TradingView
- MQL5 Documentation
- Pine Script Reference Manual

Original Code:

- FX5 MACD Divergence (MQL4)
- Alain Verleyen MQL5 Rewrite
- Community Pine Script Port

Libraries Used:

- yfinance - Yahoo Finance data
- pandas - Data manipulation
- numpy - Numerical computing
- matplotlib - Charting